Normalization:

1. Normalize the following table into 1 Normal form:

Member_Id	First_Name	Last_Name	Hobbies
101	Jayson	Mark	Cricket,Swimming,Football
102	Ram	Ganesh	Swimming,Running,Music
103	Raj	Kishore	Dancing, Singing, Running

2. Normalize the following Table into 2nd Normal Form:

Empno	Training	Training_Date	Dept
101	Oracle SQL	12-Aug-2015	TT
101	Java	21-Aug-2015	BU
102	Oracle SQL	18-Sep-2014	TT

3. Normalize the following into 3rd Normal Form:

Member_Id	First_Name	Last_Name	Sports	Fees
101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Foodball	90

Normalized Table

1NF

Member_id	First_Name	Last_Name	Hobby
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

2NF

Solution: Table ko do bhago me todna hoga

Table 1: Training_Dept (Training par aadharit)

Training	Dept
Oracle SQL	TT
Java	BU

Primary key: Training

Table 2: Emp_Training (Empno and Training par aadharit)

Empno	Training	Training_Date
101	Oracle SQL	12-Aug-2015
102	Java	21-Aug-2015
103	Oracle SQL	18-Sep-2014

Primary key: (Empno, Training)
Foreign key: Training (Yah Table 1 ki Primary key ko Reference karega)

3NF

Solution: Table ko do bhago me todna hoga

Table 1: Member (Member ki jankari)

Member_id	First_Name	Last_Name	Sports
101	Rajesh	Chand	Cricket
102	Jayson	Raj	Hockey
103	Mark	Dorson	Football

Primary key: Member_id
Foreign key: Sports (yah Table 2 ki primary key ko Reference karega)

Table 2: Sports_Fees (Sports and uski fees)

Sports	Fees
Cricket	100
Hockey	80
Football	90

Primary key: Sports

1NF, 2NF, 3NF, IMPORTANT RULs

1NF: 1) Atomic Valuses
2) No Repeating Groups
3) Unique Rows
4) Same Data Type

Example: Ek Ghar me (Cell) me ek hi admi (Value) rahega, Bheed (comma) nahi!

2NF: 1) Follow 1NF
2) Identify Composite key
3) No Partial Dependency

Example: Jodi (pair) hai to jode se bandho,akela(single) dostana nahi!

3NF: 1) Follow 2NF
2) Search Transitive Dependency
3) Move the dependency to a separate table

Example: Primary key hi malik hai, baki sab uske naukar, naukar ka naukar (chain) nahi chalega, seedha malik se hi rishta hoga!